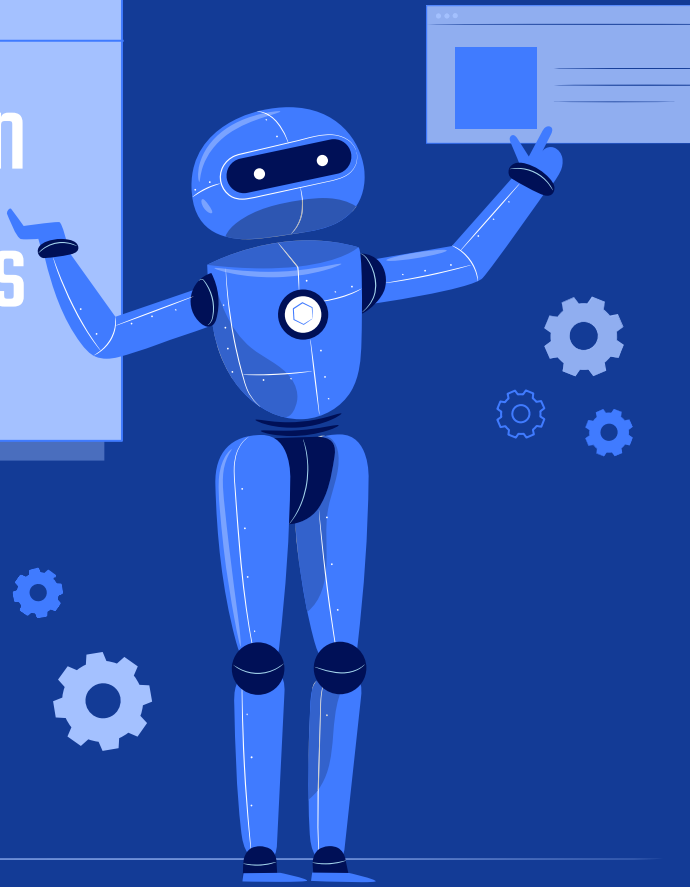
Data Extraction from Digitized Documents



Group No. - 44C

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4909 Isha Bartakke
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4963 Chhanam Thorat

Sponsored by: Hexanika Inc.
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PROBLEM

- Banks deal with large amount of documents every day & one important aspect with these documents is data entry and its storage.

- Not only the manual data entry would be hectic for the bank officials but also would lead to following problems:
  - ➢ Human error
  - ➢ Cost Incurred to train employees
  - ➢ Maintenance of hard copies



SOLUTION

- Import different types of banking documents.
- Select the data for conversion.
- Perform Optical Character Recognition (OCR) on data.
- Converts it into computer interpretable digitized text (CSV file).
- Benefits:
  - ➢ No human errors due to automation.
  - ➢ Reduction in cost of training the employees.
  - ➢ Output CSV file can be further utilized (machine readable format).

STUDY OBJECTIVES



01.

To minimize the time & cost during the manual data entry of the banking documents.

02.

To achieve higher accuracy in text identification.

03.

To increase efficiency of processing documents.

04.

To minimize the resources required during manual process

SCOPE

- The system is currently limited to the uploading of documents, extracting the data labels and its values from and storing it into the targeted CSV file to download.
- The system is restricted to English characters and numerals.
- Maintenance and storage of the output CSV file is out of scope.
- As each document upload represents an atomic operation in the module, a database management system is not required.

DOMAIN

COMPUTER VISION

For Optical Character Recognition (OCR)

APPLICATION DEVELOPMENT

Desktop-based

BANKING

Regulatory and Compliance

DELIVERABLES

SOFTWARE REQUIREMENT SPECIFICATION

01

DESIGN DOCUMENT

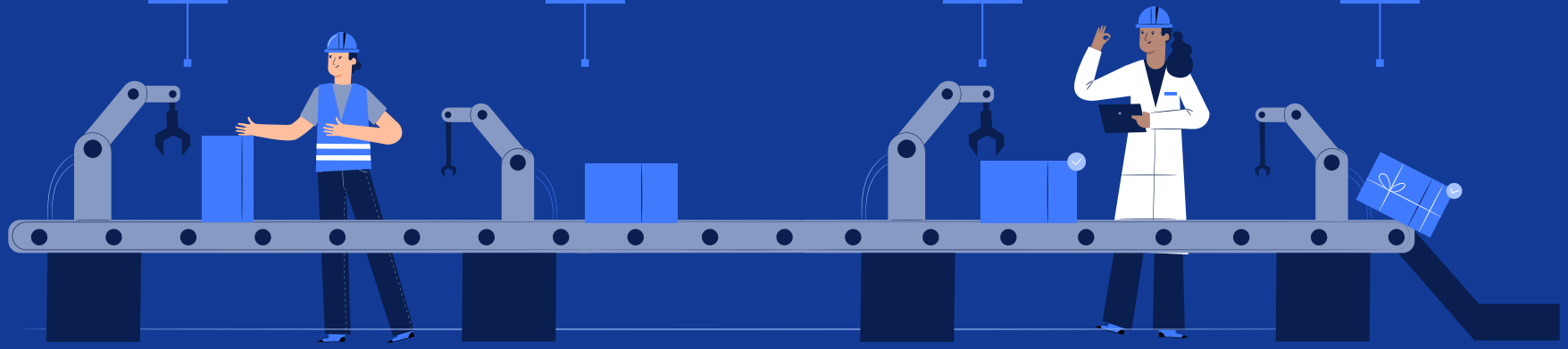
02

GUIDELINES

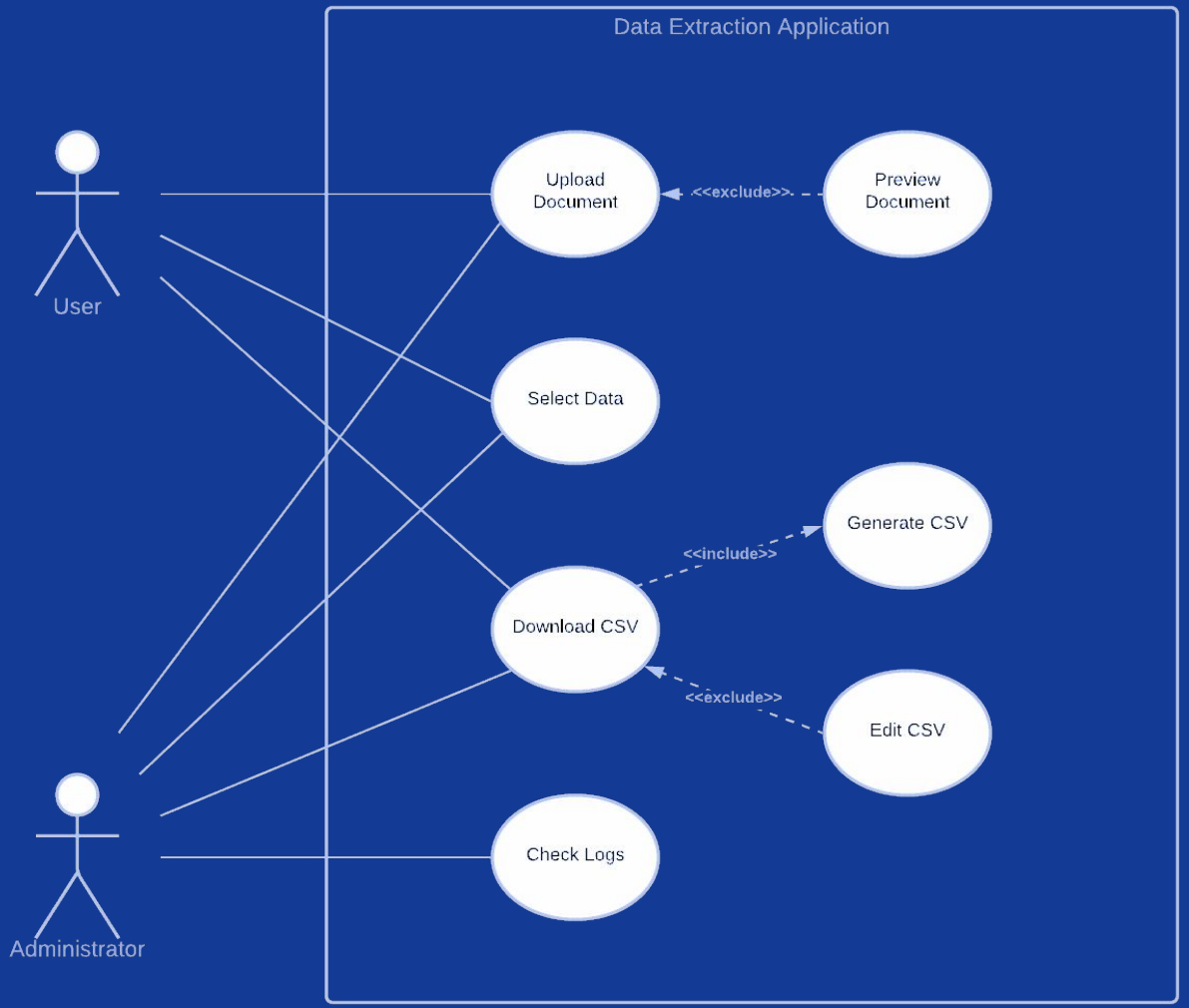
03

PRODUCT

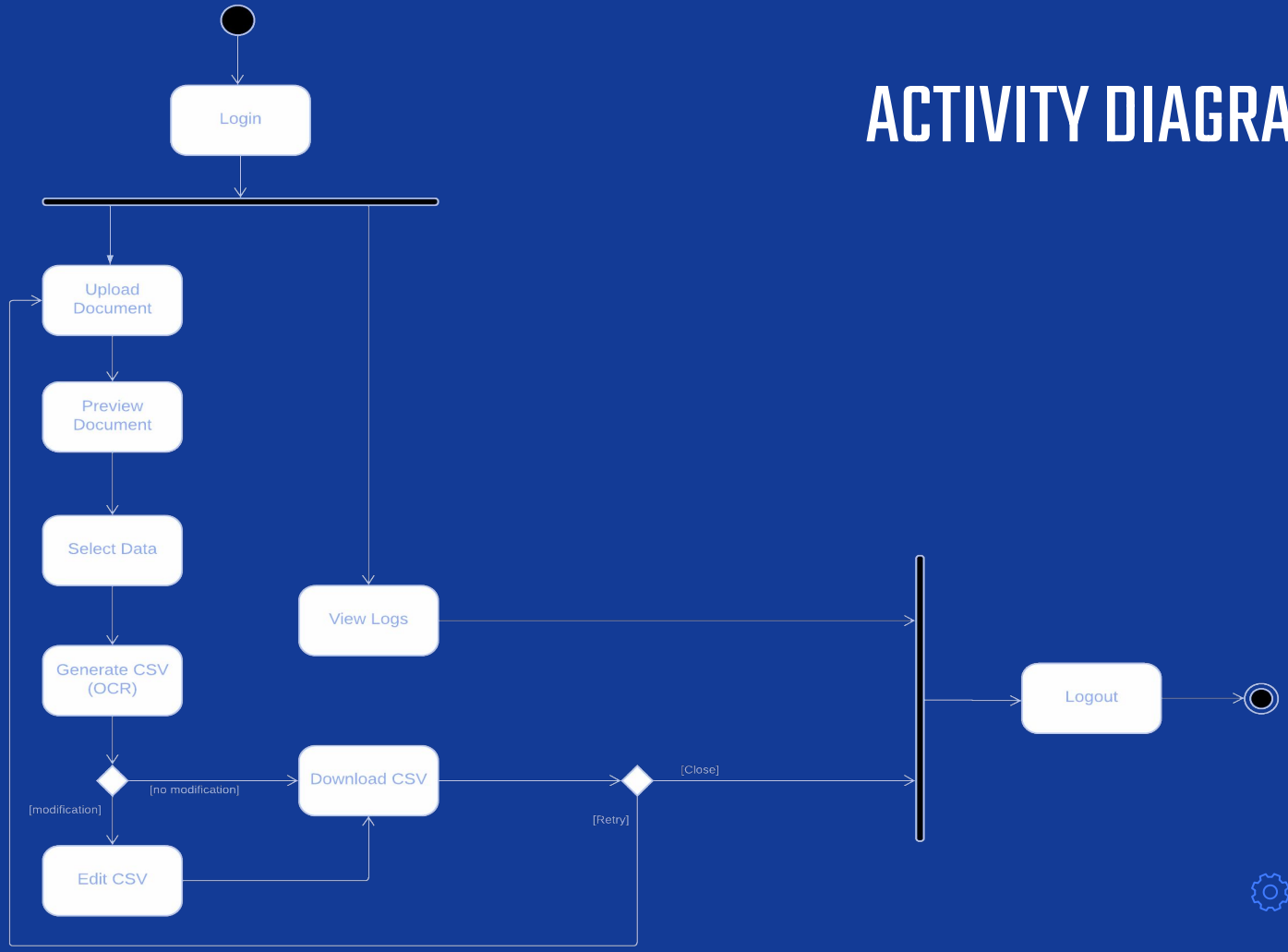
04



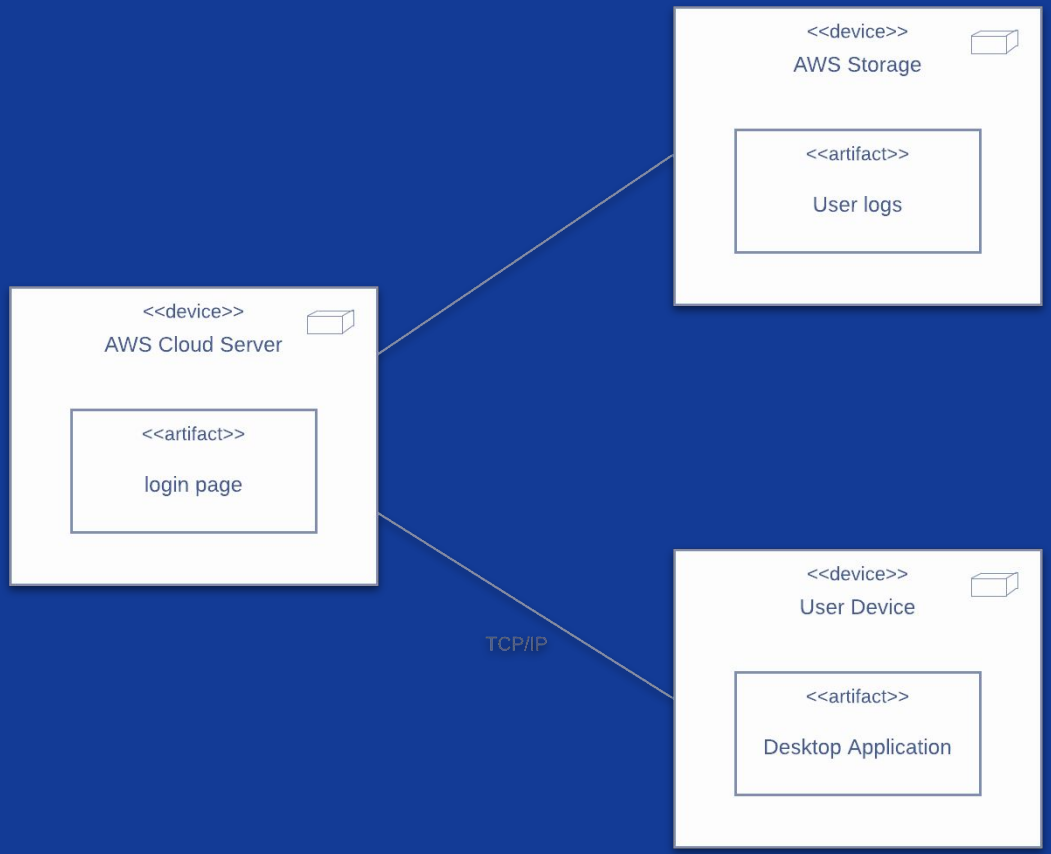
USE CASE DIAGRAM



ACTIVITY DIAGRAM



DEPLOYMENT DIAGRAM



SPECIFICATION

HARDWARE

- Machine connected in client-server architecture
  - Client: User
  - Server: AWS Cloud
- LAN connection



SOFTWARE

- Operating System - Windows, Linux
- Third Party API's - CSV file
- Commercially free Library - OCR
- Open-source tools - Design, Development & Testing

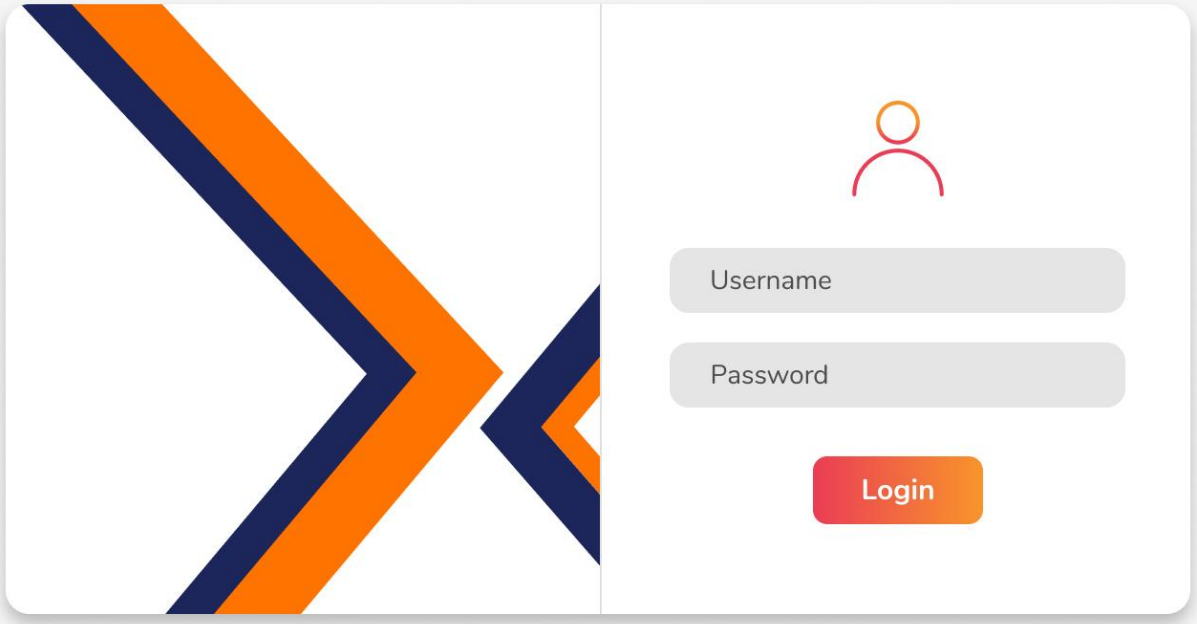
PROTOTYPE





HEXANIKA

LOGIN



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HEXANIKA

Dashboard Admin Logout

ADMIN DASHBOARD



Data Extraction Application



Logs

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HEXANIKA

Dashboard Admin Logout

DATA EXTRACTION APPLICATION



Drag and Drop File
or

Browse

- Profile.jpg — 1.2 KB / 1.2 KB
- CreditCardStatement1.jpg — 1.35 MB / 1.35 MB
- CreditCardStatement2.jpg — 271 KB / 271 KB
- TransactionHistory1.jpg — 1.5 MB / 1.5 MB
- TransactionHistory2.jpg — 1.2 MB / 1.2 MB
- TransactionHistory3.jpg — 3.5 MB / 3.5 MB

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New

Guide



BARCLAYS

Statement date 07 Dec 2012
Last statement 09 Nov 2012

Barclays Bank Account

10 Nov – 07 Dec 2012

Miss Margaret Laing
- Sort Code 20-36-16
- Account no. 10534978
- SWIFTBIC BARCGB22
- IBAN GB28 BARC 2036 1610 5349 78

MISS M LAING
33 CONRAD HOUSE
CLIFTON GROVE
DULSTON
E8 1DL

At a glance

Start balance	£370.02
Money in	£2,200.00
Money out	£2,334.74
End balance	**£235.28**

Your Barclays Bank Account statement

Current account statement

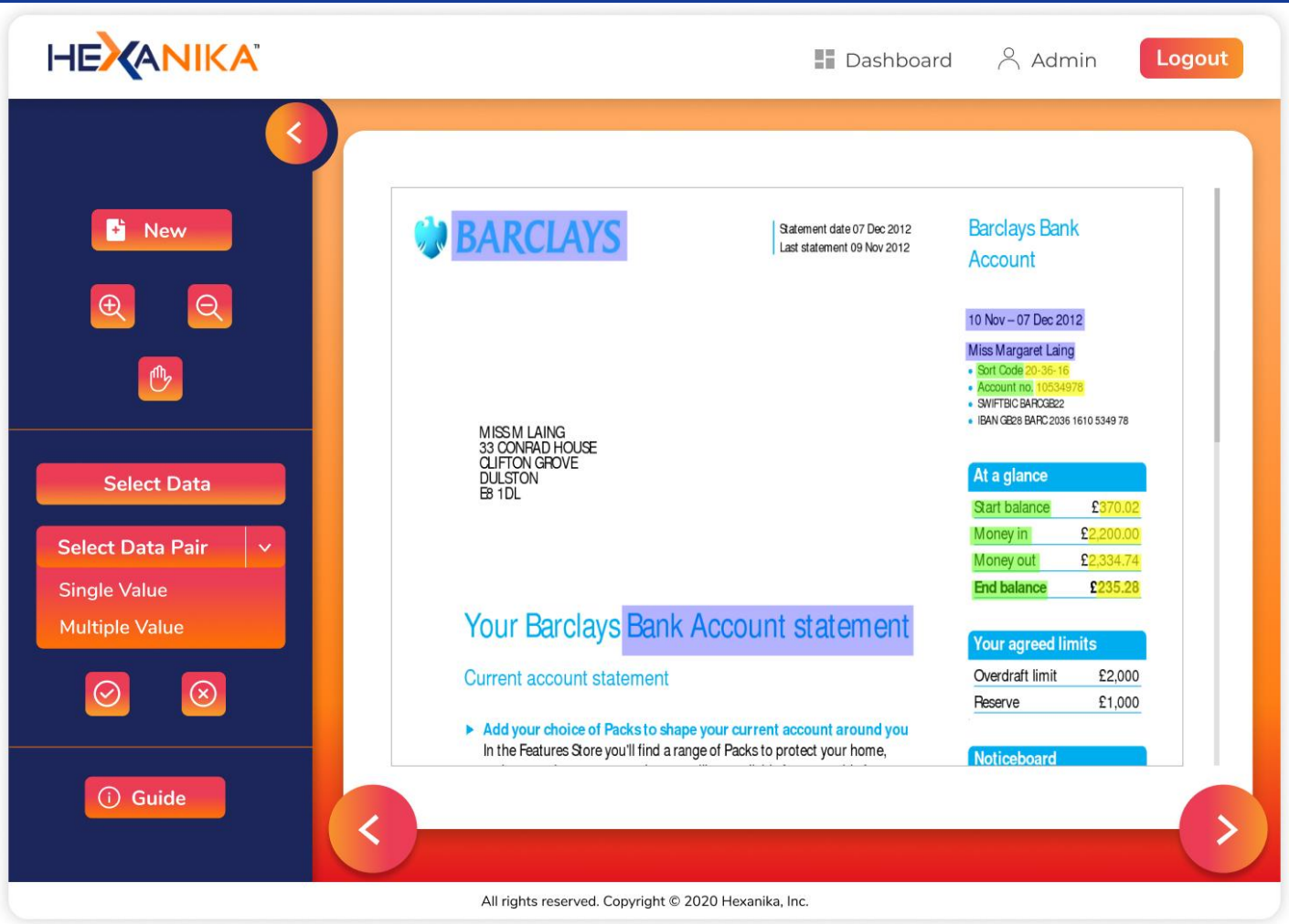
Your agreed limits

Overdraft limit	£2,000
Reserve	£1,000

▶ **Add your choice of Packs to shape your current account around you**
In the Features Store you'll find a range of Packs to protect your home,

Noticeboard

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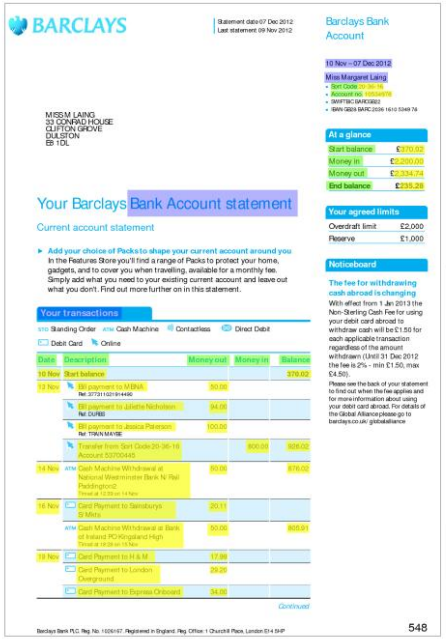
HEXANIKA

Dashboard | Admin | Logout

New

Save Changes

Cancel



BARCLAYS

Statement date 07 Dec 2012
Last statement 09 Nov 2012

Barclays Bank Account

10 Nov – 07 Dec 2012

Miss Margaret Laing

- Sort Code 20-36-16
- Account no. 10534978
- SWIFTBIC BARCGB22
- IBAN GB89 BARC 2036 1610 5349 78

MISS M LAING
33 CONRAD HOUSE
CLIFTON GROVE
DALSTON
E8 1DL

Your Barclays Bank Account statement

Current account statement

► Add your choice of Packs to shape your current account around you
In the Features Store you'll find a range of Packs to protect your home, gadgets, and to cover you when travelling, available for a monthly fee. Simply add what you need to your existing current account and leave out what you don't. Find out more further on in this statement.

At a glance

Start balance	£370.02
Money in	£2,200.00
Money out	£2,334.74
End balance	£235.28

Your agreed limits

Overdraft limit	£2,000
Reserve	£1,000

Noticeboard

The fee for withdrawing cash abroad is changing

With effect from 1 Jan 2013 the Non-Sterling Cash Fee for using your debit card abroad to withdraw cash will be £1.50 for each applicable transaction regardless of the amount withdrawn (Until 31 Dec 2012 the fee is 2% - min £1.50, max £4.50).

Please see the back of your statement to find out when the fee applies and for more information about using your debit card abroad. For details of the Global Alliance please go to barclays.co.uk/ globalalliance

Your transactions

Standing Order | Cash Machine | Contactless | Direct Debit | Debit Card | Online

Date	Description	Money out	Money in	Balance
10 Nov	Start balance			370.02
13 Nov	Bill payment to MBNA Ref: 377311021914490	50.00		
	Bill payment to Juliette Nicholson Ref: DURBI	94.00		
	Bill payment to Jessica Paterson Ref: TRAN MAYBE	100.00		
	Transfer from Sort Code 20-36-16 Account 53700445		800.00	926.02
14 Nov	ATM Cash Machine Withdrawal at National Westminster Bank N/ Rail Paddington2 Timed at 12:39 on 14 Nov	50.00		876.02
16 Nov	Card Payment to Sainsburys S/ Mkts	20.11		
	ATM Cash Machine Withdrawal at Bank of Ireland PO Kingsland High Timed at 18:28 on 15 Nov	50.00		805.91
19 Nov	Card Payment to H & M	17.99		
	Card Payment to London Overground	29.20		
	Card Payment to Express Onboard	34.00		

Continued

Barclays Bank PLC. Reg. No. 1026167. Registered in England. Reg. Office: 1 Churchill Place, London E14 5HP

548

BARCLAYS				
10 Nov - 07 Dec 2012				
Miss Margaret Liang				
Sort Code	20-36-16			
Account no.	10534978			
Start balance	370.02			
Money in	2,200.00			
Money out	2,334.74			
End balance	235.28			
Bank Account Statement				
Your transactions				
Date	Description	Money out	Money in	Balance
10-Nov	Start balance			370.02
13-Nov	Bill payment to MBNA	50		
	Bill payment to Julliette Nicholson	94		
	Bill payment to Jessica Paterson	100		
	Transfer from Sort Code 20-36-16			
	Account 53700445		800	926.02
14-Nov	Cash Machine Withdrawal at	50		876.02
	National Westminer Bank N/ Rail			
	Paddington 2			
	Timed at 12:39 on 14 Nov			
16-Nov	Card Payment to Sainsbury	20.11		
	S/ Mkts			
	Cash Machine Withdrawal at Bank	50		805.91
	of Ireland PO Kingsland High			
	Timed at 18:28 on 15 Nov			
19-Nov	Card Payment to H & M	17.99		
	Card Payment to London	29.2		
	Overground			
	Card Payment to Express Onboard	34		

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LOGS

- Input Documents
- Output CSV
- Log Files

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Logs >> Log Files

- 101_30-11-2020_11.30.21
- 101_30-11-2020_10.22.07
- 100_29-11-2020_04.54.58
- 099_29-11-2020_01.03.24
- 098_28-11-2020_12.39.35
- 097_27-11-2020_10.09.43
- 096_25-11-2020_11.45.59
- 095_11-10-2020_07.51.26

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Logs >> Log Files >> 101_30-11-2020_14.01.12.txt

```
#Login_Id: 101
#Date: Dec 03
#Timestamp: 14:01:12
#Input_Document: 101_30-11-2020_14.00.21.pdf
#Original CSV: 101_30-11-2020_14.01.12.csv
#Modified CSV: 101_30-11-2020_14.01.28.csv, 101_30-11-2020_14.30.15.csv

Dec 03 14:01:12 User modified M5 from "370.02" to "470.02".
Dec 03 14:05:13 User downloaded the generated csv.
Dec 03 14:06:19 User modified M5 from "470.02" to "485.02".
Dec 03 14:07:22 User downloaded the generated csv.
```

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STATE OF ART

- Automated Extraction of Large Scale Scanned Document Images using Google Vision OCR in Apache Hadoop Environment[IJACSA Vol.9,No.11, 2018]- This paper proposes a text extraction method of large-scale scanned document images using Google Vision OCR on the Hadoop architecture which provides 100% accuracy with twice the speed of manual data entry.
- onlineocr.net - A Free web-based OCR service which uses REST cloud API to perform OCR on the input documents.It supports 46 languages.
- Case Study on ABBYY Mobile OCR Engine -The proposed study is used for Onthego loan approval by DBS Bank, to automatically process data from documents and insert it into the correct form fields to facilitate the process of filling in loan applications.

GAP ANALYSIS

- The current system focuses on extracting data from the banking documents such as scanned images of bank statement etc and storing it in target csv file whereas the proposed system from the research paper facilitates extracting data from the university student documents and generating a text file containing that extracted data.
- The case study by dbs bank uses ABBYY mobile OCR SDK by allowing customers to fill in required data and upload imp documents and then extract the required data from it for validating the customer in order to allow on-the-go loan approval.

GAP ANALYSIS

- The onlineocr.net is an online ocr service which allows user to upload document in different formats and generate a editable text file and is used for generic purposes whereas our current project is solely used in banking sector and allows to automatically perform data entry and effectively manage files

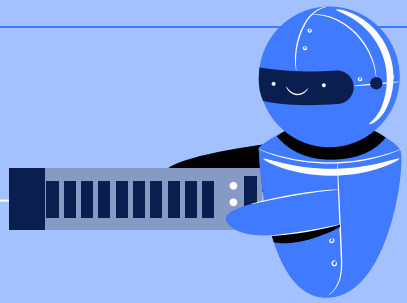
FUTURE SCOPE

HANDWRITTEN

Extend this functionality to recognize and convert text from handwritten documents

MULTILINGUAL

Improve to make the system support multilingual documents.



LEARNINGS

- Working with clients in an Agile Framework
- Designing and Prototyping gives clear vision
- Consideration for Adaptability and Integrability
- Deployment on AWS Cloud
- Adapting to the online world



REFERENCES

- [1] Rifiana Arief, Achmad Benny Mutiara, Tubagus Maulana Kusuma, Hustinawaty, Automated Extraction of Large Scale Scanned Document Images using Google Vision OCR in Apache Hadoop Environment (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 9, No. 11, 2018
- [2] DBS Bank, Case Study on ABBYY Mobile OCR Engine for On-the-go loan approval
- [3] https://www.onlineocr.net/service/about
- [4] M. Abdul Rahiman, S. Rajasree, A Detailed Study and Analysis of OCR Research in South Indian Scripts, International Conference on Advances in Recent Technologies in Communication and Computing, 2009 ISBN:978-1-4244-5104-3 Publisher: IEEE

THANK YOU

